Curriculum Vitae

Name: Liang Yue
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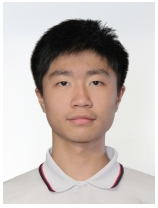

Education

Tsinghua University, Department of Computer Science and Technology
Master of Science in Computer Science Expected Graduation: June 2025
Advised by Professor Song-Hai Zhang, conducting research on "Digital 3D Indoor Scene Construction Algorithms" in the Graphics and Geometric Computing Laboratory.

Peking University, School of Information Science and Technology
Bachelor of Science in Computer Science, Double Major in Physics Graduation: June 2022
Relevant Courses: Mathematical Analysis, Computer Graphics, Artificial Intelligence, Operating Systems, etc.

Publications

- Liang Yue, Hao Tan, Junkai Huang, Shao-Kui Zhang. **A Survey on Digital 3D Indoor Scene Construction**, Journal of Image and Graphics, 2024.
- Shao-Kui Zhang, Junkai Huang, Liang Yue, Jia-Tong Zhang, Jia-Hong Liu, Yu-Kun Lai, Song-Hai Zhang. **SceneExpander: Real-Time Scene Synthesis for Interactive Floor Plan Editing**, ACM Multimedia, 2024. doi:10.1145/3664647.3680798
- Liang Yue, Shao-kui Zhang, Lin Yuan, Yi-tao Chen, Zi-rui Zhou, Song-Hai Zhang. **Synthesizing 3D Scenes via Diffusion Model that Incorporates Indoor Scene Characteristics**, ACM Multimedia, 2025. doi:10.1145/3746027.3754832
- Shao-kui Zhang, Liang Yue, Han-xi Zhu, Hao-ran Zhang, Song-Hai Zhang. **SceneCluster: Interactive Scene Synthesis by Clustering Groups of Furniture Objects**, IEEE TVCG, 2025. doi:10.1109/TVCG.2025.3634829

Internships

- Tsinghua University, College of Artificial Intelligence — Research Intern (Advisor: Miao Liu). 2025-06 – Present.
- Kunlun Skyworks — Project: World Model. 2025-07 – 2025-09. (Finetuning navigation world model; evaluation on familiar / half-familiar / unfamiliar scenes.)
- RayNeo Smart Glasses (TCL) — Project: Personal Egocentric Video Assistant. 2025-10 – Present. (Memory construction and retrieval for QA.)

Projects

Finetuning Multimodal Large Language Model with Egocentric Video Data
Enhancing the model's capability in understanding and reasoning about first-person visual information through egocentric video data. (2026-01 – till now)

Personal Egocentric Video Assistant
Developing a personal egocentric video assistant system with a focus on memory construction and retrieval for question answering. (2025-10 – till now)

Finetuning Navigation World Model with New Datasets
Finetuning the Navigation World Model using the SEKAI navigation dataset and evaluating its performance across familiar, half-familiar, and unfamiliar scenes. (2025-07 – 2025-09)

Field-based Indoor Scene Optimization Methods
Integrated features like physical constraints and layout rationality into a guiding "field" for object layout optimization. (2024-11 – 2025-04)

Semantics for Indoor Scene Layout and Dataset Augmentation
Digital indoor scenes are complex data structures; we model semantics such as "object relationships" and "functional groups" to expand indoor scene datasets for training deep generative models. (2024-07 – 2024-11)

Research Interests

- **Augmented Reality.** Next generation of Human Computer Interaction.
- **3D Computer Vision.** Mixture of Virtual and Real World.
- **Computer Graphics.** Digital Content Creation and Interaction.